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Fuentes-González and Muñoz-Durán (2017) studied the evolutionary relationships of 35 living species of the family Canidae using molecular, morphological, ecological, and other biological data. Their study showed that the relationships among canids can be understood by comparing similarities and differences among species. They also examined the positions of *Nyctereutes* and *Lycaon*, which are included in the present study. Their findings emphasize that canid species may exhibit different evolutionary relationships despite sharing common morphological characteristics. This is important in constructing a phylogenetic tree because molecular data can provide additional evidence for determining how species are related to one another. In the present study, the position of *Nyctereutes procyonoides* as an outgroup and the separation of *Lycaon pictus* from the *Canis* species can likewise be interpreted in relation to their evolutionary divergence. These findings support the use of DNA sequence data to examine the relationships and diversity of the selected Canidae species.

Similarly, Zrzavý et al. (2018) examined the relationships of living and extinct canids using different types of data, including mitochondrial and nuclear genetic markers. Their results supported the grouping of dog-like canids and showed that *Nyctereutes* and *Lycaon* have distinct positions within the Canidae family. The use of both mitochondrial and nuclear genetic information in their study demonstrates how molecular characters can reveal evolutionary relationships that may not always be evident from external morphology alone. Their findings are particularly relevant to the present analysis because the phylogenetic tree also uses DNA sequence information to compare selected canid species. The distinct placement of *Nyctereutes* and *Lycaon* further demonstrates that members of the same family can occupy different evolutionary positions, reflecting their separate evolutionary histories. This study therefore provides additional support for using genetic data to explain similarities, differences, and evolutionary relationships among canid species, which is also observed in the phylogenetic tree produced in the present study.

References

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